

TAXICAB GEOMETRY NOTES, SUMMER 2025

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1. INTRODUCTION

2. BACKGROUND AND PRELIMINARIES

Our space is

$$\mathbb{H}_T = \mathbb{H}_T^2 = \{x = (x_1, x_2) \in \mathbb{R}^2 : x_2 > 0\}$$

equipped with the hyperbolic taxicab norm

$$\|v\|_p = \frac{1}{p_2} \|v\|_T$$

where $v \in T_p(\mathbb{H}_T)$.

3. LENGTH MINIMIZING CURVES AND DISTANCE

The distance function for $\mathbb{H}_T^2$ is

$$d(p, q) = \ln \left(\frac{\mu}{p_2} \right) + \ln \left(\frac{\mu}{q_2} \right) - \frac{1}{\mu} |q_1 - p_1|$$

where $\mu = \max \{p_2, q_2, \frac{1}{2}|q_1 - p_1|\}$.

Note that if μ is equal to p_2 or q_2 , then one of the natural log terms vanishes. This corresponds to the cases where the length minimizing curves are L-shaped curves $\lambda_{p,q}$. On the other hand, if $\mu = \frac{1}{2}|q_1 - p_1|$, then the distance reduces to

$$d(p, q) = \ln \left(\frac{(q_1 - p_1)^2}{4p_2q_2} \right) + 2.$$

This corresponds to the cases where the minimizing curves are “ η ” curves. Interestingly, the length of the horizontal component of these curves is always equal to 2.

4. ISOMETRIES

□ may merge this section with the previous one, depending on how technical it gets

The set of isometries of $\mathbb{H}_T^2$ comprises horizontal translations, reflections about vertical lines, dilations centered at points on the line at infinity, and compositions of these.

□ Reflections by themselves compose to produce translations, as do dilations, so the starting set could be much smaller, at the cost of a loss of clarity.

To show that there are no isometries other than those listed above, we work through a number of technical steps. The main tools for these steps are the facts that geodesics must map to geodesics, and that the set of geodesics consists of vertical segments, horizontal segments of length two or less, L-shaped curves, and the “ η ” curves.

□ older version

Let φ be any isometry. By possibly composing with a dilation and a translation, we may suppose, without loss of generality, that $\varphi(s) = s$ where $s = (0, 1)$. Suppose that φ does not map the ray $\{x_1 = 0\}$ to itself. This means there is a point $p = (0, p_2)$ such that $\varphi(p) = q$ where, with the help of a reflection, without loss of generality, $q_1 > 0$. Moreover, $\varphi(\sigma_{s,p})$ must include a horizontal segment. By applying a new dilation and translation we may translate the point where this horizontal segment begins to s . Zooming in, we are saying that there is a triple $\{s = (0, 1), p = (0, p_2), q = (q_1, 1)\}$ such that $\varphi(s) = s$, $\varphi(p) = q$, the geodesic connecting s to p is equivalent to $\sigma_{s,p}$, and the geodesic connecting s to q is equivalent to $\sigma_{s,q}$.

□ newer version

Given two points p and q , the interval $[p, q]$ is the set of points x where the triangle equality holds:

$$[p, q] = \{x \in \mathbb{H}_T^2 : d(x, p) + d(x, q) = d(p, q)\}.$$

Lemma 4.1. *Let $\varphi : \mathbb{H}_T^2 \rightarrow \mathbb{H}_T^2$ be an isometry. Then it must map vertical segments to vertical segments and horizontal segments to horizontal segments.*

Proof. Note that all vertical line segments are intervals, and similarly all horizontal segments are unions of segments of length less than or equal to 2, and hence unions of intervals. Also, all intervals are unions of horizontal and vertical segments. Hence horizontal and vertical segments can only map to unions of horizontal and vertical segments.

(In other words, for example we can't map a horizontal segment to an angled slope or some more complicated curve.)

Suppose there exists an isometry $\varphi : \mathbb{H}_T^2 \rightarrow \mathbb{H}_T^2$ that maps a vertical line segment to a horizontal line segment. Since all vertical line segments are intervals, while a horizontal line segment is an interval if and only if its length is less than or equal to 2, we may suppose that the length of the segment is $\varepsilon \leq 2$. Let the vertical segment be $[p, q]$ with $p_2 < q_2$ so that the horizontal segment is $[\varphi(p), \varphi(q)]$. Possibly pre- and post-composing with a translation, a dilation, and a reflection, we may say that $\varphi(p) = p = s = (0, 1)$ and that $\varphi_1(q) > 0$. Now consider the point $r = (\frac{\varepsilon}{4}, 1 + \frac{\varepsilon}{2})$. Note that $[s, r]$ is the image of the L-shaped curve $\lambda_{s,r}$ which includes $[s, u] \subset [s, q]$ where $u = (0, 1 + \frac{\varepsilon}{2})$, and $[u, r]$ which, except for u , is not part of $[s, q]$.

Hence $[s, \varphi(r)]$ must include $[s, \varphi(u)] \subset [s, \varphi(q)]$ and $[\varphi(u), \varphi(r)]$ which, except for $\varphi(u)$, is not part of $[s, \varphi(q)]$. This implies that $[s, \varphi(r)]$ must also be the image of an L-shaped curve which includes a subset of $[s, \varphi(q)]$ and therefore $\varphi_2(r) < 1$.

On the other hand, $[r, q]$ is the image of an L-shaped curve and $[r, q] \cap [s, q] = \{q\}$ while $\varphi([r, q]) = [\varphi(r), \varphi(q)]$ is the image of an L-shaped curve that must include $[\varphi(u), \varphi(q)] \neq \{\varphi(q)\}$.

- this can certainly be cleaned up.
- may want to be more specific about why we can guarantee all the claimed L-shaped curves.
- not sure moving p and $\varphi(p)$ to s helps much

This is a contradiction, so it must be the case that φ maps vertical segments only to other vertical segments. Since the inverse of an isometry is also an isometry, horizontal segments must also be mapped only to horizontal segments. (If a horizontal segment mapped to a vertical one, then the inverse would map a vertical segment to a horizontal one.)

□

Next, suppose $\varphi(s) = s$. Then $\varphi((1, 1)) = (1, 1)$ or $(-1, 1)$ and by possibly composing with a reflection, we may say that without loss of generality, $\varphi((1, 1)) = (1, 1)$, and this implies that φ is the identity map on the set $\{x : x_2 = 1\}$.

The last step is to show that φ is the identity on the whole space if it is the identity on $\{x_2 = 1\}$. There seem to be a few different ways to do this:

- Look at L-shaped curves as in the lemma above.
- work in vertical strips that are not too wide.

- Show that, since vertical lines map to themselves, the only possibility other than the identity is something like $\varphi((x_1, x_2)) = \left(x_1, \frac{1}{x_2}\right)$ and this is not an isometry because it
 - fails to preserve the norm
 - fails to map L-shaped curves to L-shaped curves

Let φ be an isometry. By our lemma, we know that vertical lines are mapped to vertical lines. Suppose, without loss of generality that $\varphi(s) = s$. What are the possibilities for φ restricted to $\{x : x_1 = 0\}$? Well, it could be the identity. The only other possibility is for an inversion of some sort. Which one? Well, if I recall correctly, $(p_1, p_2) \mapsto \left(p_1, \frac{1}{p_2}\right)$ preserves the norm on each vertical line, so that should be it. It would be nice to see this using the distance function. For a point $x = (0, x_2)$, its distance from s is

$$d(x, s) = \left| \ln \left(\frac{x_2}{1} \right) \right| = |\ln(x_2)|.$$

We need $d(\varphi(x), \varphi(s)) = d(x, s)$ and since $\varphi(s) = s$ this reduces to

$$|\ln(\varphi_2(x))| = |\ln(x_2)|,$$

which in turn implies $\varphi_2(x) = x$ or $\frac{1}{x}$.

So, the only map that has a chance is $\varphi(x) = \left(x_1, \frac{1}{x_2}\right)$. But this map cannot be an isometry. Except for segments on the line $\{x : x_1 = 1\}$, lengths of horizontal segments are not preserved. Their regular taxicab length is preserved but the weight factor distorts them. For example, the distance between $\left(\frac{1}{2}, 0\right)$ and $\left(\frac{1}{2}, 1\right)$ is 2 while the distance between $\varphi\left(\left(\frac{1}{2}, 0\right)\right) = (2, 0)$ and $\varphi\left(\left(\frac{1}{2}, 1\right)\right) = (2, 1)$ is $\frac{1}{2}$.

5. GROMOV HYPERBOLICITY

6. FINAL THOUGHTS AND NEXT STEPS

7. ALINA'S SECTION

7.1. Length Calculation For Line Segments Where $p_2 \neq q_2$. We want to find $\mathcal{L}(\gamma)$ when $\gamma(t) = p + t(q - p)$ on $[0, 1]$. Breaking γ up into its coordinate points, we get $\gamma(t) = (p_1 + t(q_1 - p_1), (p_2 + t(q_2 - p_2)))$. Now using our length formula, we get

$$\begin{aligned}
\mathcal{L}(\gamma) &= \int_0^1 \left(\frac{1}{p_2 + t(q_2 - p_2)} \right) \cdot (|q_1 - p_1| + |q_2 - p_2|) dt \\
&= |q - p| \cdot \int_0^1 \frac{1}{p_2 + t(q_2 - p_2)} dt \\
&= \frac{|q - p|}{q_2 - p_2} \cdot \int_{t=0}^{t=1} \frac{1}{u} du \\
&= \frac{|q - p|}{q_2 - p_2} \cdot (\ln |u|) \Big|_{t=0}^{t=1} \\
&= \frac{|q - p|}{q_2 - p_2} \cdot (\ln |p_2 + t(q_2 - p_2)|) \Big|_0^1 \\
&= \frac{|q - p|}{q_2 - p_2} \cdot (\ln |q_2| - \ln |p_2|).
\end{aligned}$$

We used a u-substitution, where $u = p_2 + t(q_2 - p_2)$ and $\frac{du}{q_2 - p_2} = dt$. Also note that regardless of $p_2 < q_2$ or $q_2 < p_2$, the result will always be positive.

7.2. Length Calculation For Line Segments Where $p_2 = q_2$. We have $\gamma(t) = (p_1 + t(q_1 - p_1), p_2 + t(q_2 - p_2))$. We know that $p_2 = q_2$, so we can rewrite γ to be

$$\begin{aligned}
\gamma(t) &= (p_1 + t(q_1 - p_1), p_2 + t(p_2 - p_2)) \\
&= (p_1 + t(q_1 - p_1), p_2).
\end{aligned}$$

Now, we plug γ into our formula for $\mathcal{L}(\gamma)$.

$$\begin{aligned}
\mathcal{L}(\gamma) &= \int_0^1 \left(\frac{1}{p_2} \right) \cdot (|q_1 - p_1| + |0|) dt \\
&= \frac{|q_1 - p_1|}{p_2} \cdot \int_0^1 dt \\
&= \frac{d_T(p, q)}{p_2}.
\end{aligned}$$

Where, $d_T(p, q)$ is the taxicab distance between p and q , which is the same as $|q_1 - p_1|$ because we assumed that $p_2 = q_2$.

7.3. Length Calculation For Line Segments Where $p_1 = q_1$. We have $\gamma(t) = (p_1 + t(q_1 - p_1), p_2 + t(q_2 - p_2))$. We know that $p_1 = q_1$, so we can rewrite $\gamma(t)$ to be $(p_1, p_2 + t(q_2 - p_2))$. Now we plug γ into our formula for $\mathcal{L}(\gamma)$.

$$\begin{aligned}\mathcal{L}(\gamma) &= \int_0^1 \left(\frac{1}{p_2 + t(q_2 - p_2)} \right) \cdot (|0| + |q_2 - p_2|) \\ &= \frac{|q_2 - p_2|}{q_2 - p_2} \cdot \int_{t=0}^{t=1} \frac{du}{u} \\ &= \ln |q_2| - \ln |p_2| \\ &= \ln(q_2) - \ln(p_2).\end{aligned}$$

We used a u-substitution, where $u = p_2 + t(q_2 - p_2)$ and $\frac{du}{q_2 - p_2} = dt$. As $q_2 > p_2$, we know that $|q_2 - p_2| = q_2 - p_2$.

7.4. Length Minimization. We will assume that we have two curves, γ and λ between two points p , and q . We assume that $p_1 \leq q_1$, $p_2 < q_2$, $\gamma(1) = q$, and $\gamma(0) = p$. We will let σ be any line segment, so λ is a concatenation of $\sigma_{p,a}$ and $\sigma_{a,q}$. We'll let γ be any curve whose second coordinate, γ_2 , is always less than or equal to q_2 .

From our generalized line segment calculations, we know that $\mathcal{L}(\sigma_{p,a}) = \ln \left| \frac{q_2}{p_2} \right|$ and $\mathcal{L}(\sigma_{a,q}) = \frac{q_1 - p_1}{q_2}$.

For γ , we know that

$$\begin{aligned}\mathcal{L}(\gamma) &= \int_0^1 \left(\frac{1}{\gamma_2(t)} \right) \cdot (|\gamma'_1(t)| + |\gamma'_2(t)|) dt \\ &= \int_0^1 \frac{|\gamma'_1(t)|}{\gamma_2(t)} + \frac{|\gamma'_2(t)|}{\gamma_2(t)} dt \\ &= \int_0^1 \frac{|\gamma'_1(t)|}{\gamma_2(t)} dt + \int_0^1 \frac{|\gamma'_2(t)|}{\gamma_2(t)} dt\end{aligned}$$

We can evaluate each integral separately. Looking at the first integral, we know that $|\gamma'_1(t)| \geq \gamma'_1(t)$, so we'll find the minimal value by substituting in $\gamma'_1(t)$. We also know that $\gamma_2(1) = q_2$, which would give us the maximal value for γ_2 , and the minimal value for $\mathcal{L}(\gamma)$, so we'll also substitute that in.

$$\begin{aligned}
& \int_0^1 \frac{|\gamma_1'(t)|}{\gamma_2(t)} dt \\
& \geq \int_0^1 \frac{\gamma_1'(t)}{\gamma_2(t)} dt \\
& \geq \frac{1}{q_2} \cdot \int_0^1 \gamma_1'(t) dt \\
& = \frac{1}{q_2} \cdot (\gamma_1(t))|_0^1 \\
& = \frac{q_1 - p_1}{q_2} \\
& = \mathcal{L}(\sigma_{a,q})
\end{aligned}$$

Next, we'll look at the second integral. We know that $|\gamma_2'(t)| \geq \gamma_2'(t)$, so we find the minimal value by substituting in $\gamma_2'(t)$.

$$\begin{aligned}
& = \int_0^1 \frac{|\gamma_2'(t)|}{\gamma_2(t)} dt \\
& \geq \int_0^1 \frac{\gamma_2'(t)}{\gamma_2(t)} dt \\
& = \int_{t=0}^{t=1} \frac{du}{u} dt \\
& = \ln |\gamma_2(t)||_0^1 \\
& = \ln \left| \frac{q_2}{p_2} \right| \\
& = \mathcal{L}(\sigma_{p,a})
\end{aligned}$$

So, at minimum, $\mathcal{L}(\gamma) = \frac{q_1 - p_1}{q_2} + \ln \left| \frac{q_2}{p_2} \right|$, which is equal to $\mathcal{L}(\lambda)$. Therefore, we know that for any curve γ that is below λ , λ will be the minimum curve between the points p and q .

7.5. Finding All Length Minimizers. Now we'll show that if you pick a line $x = h$ where $h > q_2$, the curve created by going up from p to $x = h$, over to q_1 and then back down to q will be shorter than λ for $h = \frac{q_1 - p_1}{2} > q_2$.

So we'll call this curve η , and define η to be the concatenation of the line segments $\sigma_{p,j}, \sigma_{j,g}, \sigma_{g,q}$, where p, j, g, q are points with coordinates $p = (p_1, p_2), j = (p_1, h), g = (q_1, h), q = (q_1, q_2)$.

First, we'll show that the horizontal component of $\mathcal{L}(\eta)$ is less than or equal to the horizontal component of $\mathcal{L}(\lambda)$.

The horizontal component of $\mathcal{L}(\eta)$ is $\mathcal{L}(\sigma_{j,g})$:

$$\begin{aligned}\mathcal{L}(\sigma_{j,g}) &= \frac{g_1 - j_1}{g_2} \\ &= \frac{q_1 - p_1}{h} \\ &< \frac{q_1 - p_1}{q_2}.\end{aligned}$$

So, the horizontal component of $\mathcal{L}(\eta)$ is strictly less than the horizontal component of $\mathcal{L}(\lambda)$.

Now we'll see if the vertical components of $\mathcal{L}(\eta)$ are less than or equal to the vertical component of $\mathcal{L}(\lambda)$.

The vertical components of $\mathcal{L}(\eta)$ are $\mathcal{L}(\sigma_{p,j}) * \mathcal{L}(\sigma_{g,q})$:

$$\begin{aligned}\mathcal{L}(\sigma_{p,j}) &= \ln \left| \frac{h}{p_2} \right| \\ \mathcal{L}(\sigma_{g,q}) &= -\ln \left| \frac{q_2}{h} \right| \\ \mathcal{L}(\sigma_{p,j}) * \mathcal{L}(\sigma_{g,q}) &= \ln \left| \frac{h}{p_2} \right| - \ln \left| \frac{q_2}{h} \right| \\ &= \ln \left| \frac{h}{p_2} \div \frac{q_2}{h} \right| \\ &= \ln \left| \frac{h}{p_2} \cdot \frac{h}{q_2} \right| \\ &= \ln \left| \frac{h^2}{p_2 q_2} \right| \\ &> \ln \left| \frac{q_2}{p_2} \right|.\end{aligned}$$

So, the vertical components of $\mathcal{L}(\eta)$ are greater than the vertical component of $\mathcal{L}(\lambda)$.

Therefore, we don't know if $\mathcal{L}(\eta) \leq \mathcal{L}(\lambda)$.

So, we'll find the optimal h value for this calculation is $\frac{q_1 - p_1}{2}$.

First we take the derivative of $\mathcal{L}(\eta)$ with respect to h :

$$\begin{aligned}\frac{d}{dh}(\mathcal{L}(\eta)) &= -\frac{|q_1 - p_1|}{h^2} + \frac{2}{h} \\ &= \frac{2h - |q_1 - p_1|}{h^2}.\end{aligned}$$

Then we set the derivative equal to zero:

$$\begin{aligned}\frac{2h - |q_1 - p_1|}{h^2} &= 0 \\ 2h - |q_1 - p_1| &= 0 \\ 2h &= |q_1 - p_1| \\ h &= \frac{|q_1 - p_1|}{2}.\end{aligned}$$

So, the critical value for h is $\frac{|q_1 - p_1|}{2}$.

Next, we'll show that this h value is a minimum.

To do this, we'll do the second derivative test.

First, we'll find the second derivative:

$$\begin{aligned}\frac{d}{dh} \left(\frac{1}{h^2} \cdot (2h - |q_1 - p_1|) \right) &= \left(-\frac{2}{h^3} \cdot (2h - |q_1 - p_1|) \right) + \left(2 \cdot \frac{1}{h^2} \right) \\ &= \frac{2}{h^2} - \frac{4h}{h^3} + \frac{2 \cdot |q_1 - p_1|}{h^3} \\ &= \left(\frac{2}{h^3} \cdot |q_1 - p_1| \right) - \frac{2}{h^2}.\end{aligned}$$

Now we plug in h :

$$\begin{aligned}
& \left(\frac{2}{\left(\frac{|q_1 - p_1|}{2}\right)^3} \cdot |q_1 - p_1| \right) - \frac{2}{\left(\frac{|q_1 - p_1|}{2}\right)^2} \\
&= \left((2 \cdot 2^3) \cdot \frac{|q_1 - p_1|}{|q_1 - p_1|^3} \right) - \frac{2 \cdot 2^2}{|q_1 - p_1|^2} \\
&= \frac{16}{|q_1 - p_1|^2} - \frac{8}{|q_1 - p_1|^2} \\
&= \frac{8}{|q_1 - p_1|^2} \\
&> 0.
\end{aligned}$$

Because the second derivative is greater than 0, we know that h is a minimum. Therefore, $\frac{|q_1 - p_1|}{2}$ is the optimal value for h . We know that any value around h will mean that $\mathcal{L}(\eta) < \mathcal{L}(\lambda)$, and at h $\mathcal{L}(\eta)$ is the shortest path.

7.6. Length Minimization For Curves Below η and Above λ .

Let's assume we have a curve γ who's second coordinate γ_2 achieves a maximum value at $t = t_0$, and there's a point $g = \gamma(t_0)$. We will also assume that some η_2 's are equal to g .

Since η is composed of three concatenated line segments, and γ_2 is equal to η_2 at one point g , we will break up η into two λ curves, $\lambda_{p,g}$ and $\lambda_{q,g}$. We will also state that γ is equivalent to a concatenation of two curves, $\gamma_{p,g}$ and $\gamma_{g,q}$.

So, we can evaluate $\mathcal{L}(\gamma_{p,g})$ in relation to $\mathcal{L}(\lambda_{p,g})$ and $\mathcal{L}(\gamma_{g,q})$ in relation to $\mathcal{L}(\lambda_{q,g})$.

As we showed in section 2.4, any curve γ whose second coordinate is less than or equal to the second coordinate of λ for all γ_2, λ_2 , the length of λ is always less than or equal to the length of γ .

Therefore, $\mathcal{L}(\lambda_{p,g}) \leq \mathcal{L}(\gamma_{p,g})$ and $\mathcal{L}(\lambda_{q,g}) \leq \mathcal{L}(\gamma_{g,q})$. So, $\mathcal{L}(\eta) \leq \mathcal{L}(\gamma)$.

7.7. The Distance Function of $\mathbb{H}_{0,T}^2$. Now that we have our length minimizers, we need to come up with a distance function that encompasses both λ and η .

We know that η only works when $\frac{|q_1 - p_1|}{2} > q_2$, and for every other case we need to use λ . So we need to find a general way to write out the horizontal and vertical components into one equation.

First we'll look at the horizontal components of λ and η :

$$\begin{aligned}\lambda &= \frac{|q_1 - p_1|}{\max\{p_2, q_2\}} \\ \eta &= \frac{|q_1 - p_1|}{|q_1 - p_1| \cdot \frac{1}{2}} \\ &= \frac{|q_1 - p_1|}{\max\{p_2, q_2, h\}} \\ &= \frac{|q_1 - p_1|}{\mu}\end{aligned}$$

We'll define μ to be $\max\{p_2, q_2, h\}$. Observe that when h is the max of the set, the horizontal component becomes 2.

Now we have an easy way to write out the horizontal components of λ and η as one.

Second, we'll look at the vertical components of λ and η :

$$\begin{aligned}\lambda &= \left| \ln \left(\frac{q_2}{p_2} \right) \right| \\ \eta &= \ln \left| \frac{h^2}{q_2 \cdot p_2} \right| \\ &= \ln \left| \frac{\mu}{p_2} \right| + \ln \left| \frac{\mu}{q_2} \right| \\ &= \ln \left| \frac{\mu^2}{p_2 \cdot q_2} \right|\end{aligned}$$

We take μ to still be $\max\{p_2, q_2, h\}$. Observe that when the max of the set is p_2 or q_2 , we obtain the equation for λ , and when the max is h , we obtain the equation for η .

Again, we've obtained an easy way to write out the vertical components of λ and η as one.

So,

$$d(p, q)_{\mathbb{H}_{0,T}^2} = \ln \frac{\mu^2}{p_2 \cdot q_2} + \frac{|q_1 - p_1|}{\mu}.$$

7.8. A Generalized Norm for Different Lines at Infinity. Now we're going to look at different lines at infinity to see how that changes the space. We assume that $0 \leq m \leq 1$ describes the slope for every possible line at infinity. To do that we need a norm that will work for every line at infinity.

When the line at infinity has a slope of 0, we know that the norm is $\frac{\|v\|_T}{p_2}$, where $\|v\|_T$ is the regular taxicab norm.

Since we have a line at infinity with a slope of m , we need a different weight on the taxicab norm. We observed that the vertical distance, the shortest distance to the line at infinity, from a point p to the line at infinity with a slope m , is $p_2 - mp_1$. So, the weight we need for our norm is $\frac{1}{p_2 - mp_1}$.

Therefore the norm is $\frac{\|v\|_T}{p_2 - mp_1}$ for any line at infinity.

7.9. Defining Horizontal and Vertical Lines for All Lines at Infinity. Since we're moving on to looking at lines at infinity other than $x_2 = 0$, we need a clear definition of what horizontal and vertical lines are.

We will assume that instead of a horizontal line being parallel to the x-axis, a horizontal line is parallel to the line at infinity. We will also assume that instead of a vertical line being perpendicular to the x-axis or parallel to the y-axis, a vertical line is perpendicular to the line at infinity.

Sometimes we will also need to discuss lines that are horizontal or vertical with respect to the axes, and we will denote these as coordinate-horizontal and coordinate-vertical.

7.10. Horizontal Translations as Isometries for a Line at Infinity With a Slope of m , where $0 < m < 1$. We want to show that a map Φ that translates a point p across a horizontal line is an isometry of $\mathbb{H}_{m,T}^2$. We will call a the amount that p is being moved across the horizontal line.

So,

$$\begin{aligned}\Phi(p) &= p + (a, ma) \\ &= (p_1 + a, p_2 + ma).\end{aligned}$$

The map Φ has a pushforward Φ_* , represented by the Jacobian matrix:

$$\begin{aligned}\Phi_* &= \begin{bmatrix} \frac{\delta\Phi_1}{\delta p_1} & \frac{\delta\Phi_1}{\delta p_2} \\ \frac{\delta\Phi_2}{\delta p_1} & \frac{\delta\Phi_2}{\delta p_2} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\end{aligned}$$

Now that we have our Jacobian matrix, we can confirm that the norm of $\Phi_*(v)$ centered at a point $\Phi(p)$ is the same as the norm of a vector centered at our original point p :

$$\begin{aligned}\|\Phi_*(v)\|_{m, \Phi(p)} &= \frac{|\Phi_*(v_1)| + |\Phi_*(v_2)|}{\Phi_2(p) - m\Phi_1(p)} \\ &= \frac{|v_1| + |v_2|}{p_2 + ma - m(p_1 + a)} \\ &= \frac{|v_1| + |v_2|}{p_2 - mp_1} \\ &= \|v\|_{m, p}.\end{aligned}$$

Since we got out the norm of a vector centered at our original point p , horizontal translations are an isometry of $\mathbb{H}_{m, T}^2$.

7.11. Dilations Centered at the Origin as Isometries for a Line at Infinity With a Slope of m , where $0 < m < 1$. We want to show that a map Φ that dilates a point p from the origin is an isometry of $\mathbb{H}_{m, T}^2$. We will call k the amount that p is being dilated.

So,

$$\begin{aligned}\Phi(p) &= k \cdot p \\ &= (kp_1, kp_2).\end{aligned}$$

This map also has a pushforward that we'll call Φ_* , represented by the

Jacobian matrix:

$$\Phi_* = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$$

Now that we have the Jacobian matrix, we can confirm that the norm of $\Phi_*(v)$ centered at a point $\Phi(p)$ is the same as the norm of a vector centered at our original point p :

$$\begin{aligned} \|\Phi_*(v)\|_{m,\Phi(p)} &= \frac{|\Phi_*(v_1)| + |\Phi_*(v_2)|}{\Phi_2(p) - m\Phi_1(p)} \\ &= \frac{k|v_1| + k|v_2|}{kp_2 - mkp_1} \\ &= \frac{k \cdot (|v_1| + |v_2|)}{k \cdot (p_2 - mp_1)} \\ &= \frac{k}{k} \cdot \frac{|v_1| + |v_2|}{p_2 - mp_1} \\ &= \frac{|v_1| + |v_2|}{p_2 - mp_1} \\ &= \|v\|_{m,p}. \end{aligned}$$

Since we got out the norm of a vector centered at our original point p , even dillations centered at the origin are an isometry of $\mathbb{H}_{m,T}^2$.

7.12. General Length Calculation for Curves in $\mathbb{H}_{m,T}^2$, where $p \neq q$. Now we will look at length calculations for curves made up of a concatenation of line segments. We will call this curve γ , with coordinates of $(p_1 + t(q_1 - p_1), p_2 + t(q_2 - p_2))$.

Now we plug this into the length function and find the formula:

$$\begin{aligned}
\mathcal{L}(\gamma) &= \int_0^1 \frac{|\gamma'_1(t)| + |\gamma'_2(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\
&= \int_0^1 \frac{|q_1 - p_1| + |q_2 - p_2|}{p_2 + t(q_2 - p_2) - m(p_1 + t(q_1 - p_1))} dt \\
&= |q - p| \int_0^1 \frac{1}{p_2 - mp_1 + t(q_2 - p_2) - mt(q_1 - p_1)} dt \\
&= \frac{|q - p|}{q_2 - mq_1 + mp_1 - p_2} \cdot \int_{t=0}^{t=1} \frac{1}{u} du \\
&= \frac{|q - p|}{q_2 - mq_1 + mp_1 - p_2} \cdot \ln |p_2 - mp_1 + t(q_2 - p_2) - mt(q_1 - p_1)| \Big|_0^1 \\
&= \frac{|q - p|}{q_2 - mq_1 + mp_1 - p_2} \cdot (\ln |q_2 - mq_1| - \ln |p_2 - mp_1|) \\
&= \frac{|q - p| \cdot \ln \left| \frac{q_2 - mq_1}{p_2 - mp_1} \right|}{q_2 - p_2 + m(p_1 - q_1)}.
\end{aligned}$$

We used a u -substitution, where $u = p_2 - mp_1 + t(q_2 - p_2) - mt(q_1 - p_1)$.

7.13. Length Formula for Coordinate-Horizontal Line Segments in $\mathbb{H}_{m,T}^2$. Here we will assume that $p_2 = q_2$. Then observe that when we plug this condition into the general formula, we get out:

$$\frac{1}{m} \cdot \ln \left| \frac{p_2 - mp_1}{p_2 - mq_1} \right|.$$

7.14. Length Formula for Coordinate-Vertical Line Segments in $\mathbb{H}_{m,T}^2$. Here we will assume that $p_1 = q_1$. Then observe that when we plug this condition into the general formula, we get out:

$$\ln \left| \frac{q_2 - mp_1}{p_2 - mp_1} \right|.$$

7.15. Length Formula for Horizontal Line Segments in $\mathbb{H}_{m,T}^2$. Here we will assume that $q_2 = p_2 + m(q_1 - p_1)$. Observe that when we plug this condition into the general formula, we get out:

$$\frac{(q_1 - p_1) \cdot (m + 1)}{p_2 - mp_1}.$$

7.16. Length Formula for Vertical Line Segments in $\mathbb{H}_{m,T}^2$. Here we will assume that $q_2 = p_2 + \frac{q_1 - p_1}{m}$. Observe that when we plug this condition into the general formula, we get out:

$$\frac{1+m}{1+m^2} \cdot \ln \left| \frac{p_2 - mp_1}{p_2 - q_1(\frac{1}{m} + m) + \frac{p_1}{m}} \right|.$$

7.17. Comparing the Length of a Curve λ to the Length of a General Curve γ . Here we're going to compare the lengths of two curves λ and γ . λ is the concatenation of two line segments $\sigma_{p,a}$ and $\sigma_{a,q}$, where $\sigma_{a,q}$ is a coordinate-horizontal line segment from point p to point a , and $\sigma_{p,a}$ is a coordinate-vertical line segment from point a to point q . Point a has coordinates (p_1, q_2) . γ is any doubly monotonic curve who's second coordinate γ_2 is less than or equal to q_2 at every point.

By plugging the coordinates of a into the length formulas for coordinate-horizontal and coordinate-vertical lines, we get out the lengths of $\sigma_{p,a}$ and $\sigma_{a,q}$:

$$\begin{aligned} \sigma_{p,a} &= \ln \left| \frac{q_2 - mp_1}{p_2 - mp_1} \right| \\ \sigma_{a,q} &= \frac{1}{m} \cdot \ln \left| \frac{q_2 - mp_1}{q_2 - mq_1} \right|. \end{aligned}$$

Now, we'll compute the length of γ :

$$\begin{aligned} \mathcal{L}(\gamma) &= \int_0^1 \frac{|\gamma'_1(t)| + |\gamma'_2(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\ &= \int_0^1 \frac{|\gamma'_1(t)| + m|\gamma'_1(t)| + |\gamma'_2(t)| - m|\gamma'_1(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\ &= \int_0^1 \frac{|\gamma'_1(t)| + m|\gamma'_1(t)|}{\gamma_2(t) - m\gamma_1(t)} dt + \int_0^1 \frac{|\gamma'_2(t)| - m|\gamma'_1(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \end{aligned}$$

First, we'll look at the coordinate-horizontal component of γ :

$$\begin{aligned}
& \int_0^1 \frac{|\gamma_1'(t)| + m|\gamma_1'(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\
&= (1+m) \cdot \int_0^1 \frac{|\gamma_1'(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\
&= (1+m) \cdot \int_0^1 \frac{\gamma_1'(t)}{\gamma_2(t) - m\gamma_1(t)} dt \\
&\geq (1+m) \cdot \int_0^1 \frac{\gamma_1'(t)}{q_2 - m\gamma_1(t)} dt \\
&= \frac{1+m}{-m} \cdot \int_{t=0}^{t=1} \frac{1}{u} du \\
&= \frac{1+m}{-m} \cdot \ln \|q_2 - m\gamma_1(t)\|_0^1 \\
&= \frac{1+m}{-m} \cdot \ln \left| \frac{q_2 - mq_1}{q_2 - mp_1} \right| \\
&= \frac{1+m}{m} \cdot \ln \left| \frac{q_2 - mp_1}{q_2 - mq_1} \right|
\end{aligned}$$

We used a u -substitution, where $u = q_2 - m\gamma_1(t)$. This length is greater than the length of $\sigma_{a,q}$.

Next, we'll look at the coordinate-vertical component of γ :

$$\begin{aligned}
& \int_0^1 \frac{|\gamma_2'(t)| - m|\gamma_1'(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\
&= \int_0^1 \frac{\gamma_2'(t) - m\gamma_1'(t)}{\gamma_2(t) - m\gamma_1(t)} dt \\
&= \int_0^1 \frac{1}{u} du \\
&= \ln \|q_2(t) - m\gamma_1(t)\|_0^1 \\
&= \ln |q_2 - mq_1| - \ln |p_2 - mp_1|
\end{aligned}$$

Let's compare this length to the length of $\sigma_{p,a}$:

$$\begin{aligned}
& \sigma_{p,a} \cdot \ln |q_2 - mq_1| - \ln |p_2 - mp_1| \\
& \ln |q_2 - mp_1| - \ln |p_2 - mp_1| \cdot \ln |q_2 - mq_1| - \ln |p_2 - mp_1| \\
& \ln |q_2 - mp_1| \cdot \ln |q_2 - mq_1| \\
& q_2 - mp_1 \cdot q_2 - mq_1 \\
& -mp_1 > -mq_1 \\
& mp_1 < mq_1
\end{aligned}$$

So, the length of λ is less than the length of γ .

7.18. $\mathbb{H}_m^2$ Length Minimization. We will show that we have a length minimizer λ made up of a concatenation of coordinate-horizontal and coordinate-vertical line segments. We have two cases here, one where $p \leq q$, and one where $p_1 \geq q_1$, $p_2 \leq q_2$.

First we'll look at the case where $p \leq q$:

We have points p and q with coordinates $p = (p_1, p_2)$ and $q = (q_1, q_2)$. We have a point a that allows us to break up the curve λ into two line segments, which has coordinates $a = (p_1, q_2)$. Our line segments are $\sigma_{p,a}$ which is a coordinate-vertical line segment, and $\sigma_{a,q}$ which is a coordinate-horizontal line segment.

We know that the length of $\sigma_{p,a} = \ln \left| \frac{q_2 - mp_1}{p_2 - mp_1} \right|$ and the length of $\sigma_{a,q} = \frac{1}{m} \cdot \ln \left| \frac{q_2 - mp_1}{q_2 - mq_1} \right|$.

We also have a point t_0 which we'll define to be the last time $\gamma_1(t)$ is equal to p_1 . We know that this point will exist because of the intermediate value theorem. We know that $\gamma(0) = p$ and $\gamma(1) = q$. We will also call q_2 the greatest value that γ_2 ever reaches and bound γ at that point.

Now, we'll split γ into a concatenation of γ_{0,t_0} and $\gamma_{t_0,1}$, where γ_{0,t_0} is the portion of γ that lives between the point p and the point t_0 and $\gamma_{t_0,1}$ is the portion of γ that lives between the point t_0 and the point q .

7.18.1. First we'll look at γ_{0,t_0} :

$$\begin{aligned}
\mathcal{L}(\gamma_{0,t_0}) &= \int_0^1 \frac{|\gamma'_1(t)| - m|\gamma'_1(t)| + |\gamma'_2(t)| + m|\gamma'_1(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\
&= \int_0^1 \frac{(1-m) \cdot |\gamma'_1(t)|}{\gamma_2(t) - m\gamma_1(t)} dt + \int_0^1 \frac{|\gamma'_2(t) - m\gamma'_1(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\
&\geq (1-m) \cdot \int_0^1 \frac{|\gamma'_1(t)|}{q_2 - m\gamma_1(t)} dt + \left| \int_{t=0}^{t=1} \frac{1}{u} du \right| \\
&\geq (1-m) \cdot \left| \int_0^1 \frac{\gamma'_1(t)}{q_2 - m\gamma_1} dt \right| + \left| \ln \left(\frac{q_2 - mq_1}{p_2 - mp_1} \right) \right| \\
&= (1-m) \cdot \left| -\frac{1}{m} \int_{t=0}^{t=1} \frac{1}{u} du \right| + |\mathcal{L}(\sigma_{p,t_0})| \\
&= \frac{1-m}{m} \left| \ln \left(\frac{q_2 - mq_1}{q_2 - mp_1} \right) \right| + \mathcal{L}(\sigma_{p,t_0})
\end{aligned}$$

We use the triangle inequality to substitute in $|\gamma'_2(t) - m\gamma'_1(t)|$ is less than or equal to $|\gamma'_2(t)| + m|\gamma'_1(t)|$. Then we substituted in q_2 for γ_2 , as γ_2 is bounded by q_2 . We also used a u -substitution, where $u = \gamma_2(t) - m\gamma_1(t)$ to reduce the second integral down to just $\frac{1}{u}$. We also moved the absolute value bars to encompass the whole integrals because the absolute value of the whole term is less than or equal to the absolute value of just the γ terms plugged into the whole thing. Finally, we can get rid of the absolute value bars around the length, as lengths are always positive.

Observe that as we minimize γ , the horizontal term approaches zero, leaving just the vertical path between the points to be the shortest path.

7.18.2. Now we'll look at $\gamma_{t_0,1}$:

8. KEVIN'S SECTION

CONTINUITY OF THE LENGTH FUNCTION

We want to show that the piecewise formula for $\mathcal{L}(r)$ is continuous as $p_2 = q_2$. That is, we want to verify that:

$$\lim_{q_2 \rightarrow p_2} \left(\frac{|q_1 - p_1|}{q_2 - p_2} \ln \left(\frac{q_2}{p_2} \right) \right) = \frac{|q_1 - p_1|}{p_2}$$

Case 1: $p_2 = q_2$. Here the formula simplifies to:

$$\mathcal{L}(r) = \frac{|q_1 - p_1|}{p_2} = \frac{x}{a}$$

Case 2: $p_2 \neq q_2$. Let us denote:

$$x := |q_1 - p_1|, \quad a := p_2, \quad \varepsilon := q_2 - p_2$$

Then as $\varepsilon \rightarrow 0$, we can write:

$$\mathcal{L}(r) = \frac{x}{\varepsilon} \ln \left(\frac{a + \varepsilon}{a} \right) = \frac{x}{\varepsilon} \ln \left(1 + \frac{\varepsilon}{a} \right)$$

Using the Taylor expansion:

$$\ln(1 + k) = k - \frac{k^2}{2} + \frac{k^3}{3} - \dots$$

we approximate:

$$\ln \left(1 + \frac{\varepsilon}{a} \right) = \frac{\varepsilon}{a}$$

Hence:

$$\mathcal{L}(r) = \frac{x}{\varepsilon} \cdot \frac{\varepsilon}{a} = \frac{x}{a}$$

Conclusion. Thus,

$$\lim_{q_2 \rightarrow p_2} \left(\frac{|q_1 - p_1|}{q_2 - p_2} \ln \left(\frac{q_2}{p_2} \right) \right) = \frac{x}{a}$$

which matches the formula in Case 2. Therefore, the length function $\mathcal{L}(r)$ is continuous as $q_2 \rightarrow p_2$.

Second Derivative and Inflection Point:

The second derivative is

$$\frac{d^2 \mathcal{L}}{dh^2} = -\frac{2}{(q_2 + h)^2} + \frac{2|q_1 - p_1|}{(q_2 + h)^3} = \frac{2(|q_1 - p_1| - (q_2 + h))}{(q_2 + h)^3}$$

Critical Point:

$$h^* = \frac{|q_1 - p_1|}{2} - q_2$$

$$q_2 + h^* = \frac{|q_1 - p_1|}{2}$$

$$\left. \frac{d^2 \mathcal{L}}{dh^2} \right|_{h^*} = \frac{8}{|q_1 - p_1|^2}$$

which is always positive ($q_1 \neq p_1$), so h^* is a local minimum.

Inflection Point: The inflection point occurs when

$$|q_1 - p_1| - (q_2 + h) = 0 \implies h_{\text{inf}} = |q_1 - p_1| - q_2$$

For $h < h_{\inf}$, $\frac{d^2\mathcal{L}}{dh^2} > 0$: $\mathcal{L}$ is concave up.

For $h > h_{\inf}$, $\frac{d^2\mathcal{L}}{dh^2} < 0$: $\mathcal{L}$ is concave down.

In this plot, we set $|q_1 - p_1| = 4$, $q_2 = 2$, $p_2 = 1$. The vertical red dashed line at $h_{\inf} = 2$ is the inflection point, running from $g(h) = 2.7$ to $g(h) = 4.2$.

GENERALIZED LENGTH CALCULATION FOR LINE SEGMENTS

Let $p = (p_1, p_2)$, $q = (q_1, q_2)$. Define a parametric path:

$$\gamma(t) = p + t(q - p), \quad t \in [0, 1]$$

Then,

$$\gamma'(t) = q - p = (q_1 - p_1, q_2 - p_2)$$

Define the weight function (depending on height):

$$\delta(t) = p_2 + t(q_2 - p_2)$$

The length functional becomes:

$$\mathcal{L} = \int_0^1 \frac{|\gamma'_1(t)| + |\gamma'_2(t)|}{\delta(t)} dt = \int_0^1 \frac{|q_1 - p_1| + |q_2 - p_2|}{p_2 + t(q_2 - p_2)} dt$$

Assume $q_2 \neq p_2$. Make substitution:

$$u = p_2 + t(q_2 - p_2), \quad du = (q_2 - p_2) dt$$

Then:

$$\mathcal{L} = \int_{p_2}^{q_2} \frac{|q_1 - p_1| + |q_2 - p_2|}{u(q_2 - p_2)} du = \left(\frac{|q_1 - p_1| + |q_2 - p_2|}{q_2 - p_2} \right) \int_{p_2}^{q_2} \frac{1}{u} du$$

So finally:

$$\boxed{\mathcal{L} = \frac{d_t(p, q)}{q_2 - p_2} \ln \left(\frac{q_2}{p_2} \right)}$$

where $d_t(p, q) = |q_1 - p_1| + |q_2 - p_2|$ is the taxicab distance.

PROOFS THAT DILATION, TRANSLATION, AND REFLECTION ARE ISOMETRIES

Let $A = \mathbb{H}_T^2$. We show that dilation, translation, and reflection maps are isometries of A by proving each is a bijection and translation, and reflection maps preserve distances.

1. Dilation Map. The dilation map about the origin is defined as

$$\delta_k(p) = k \cdot p,$$

for some $k > 0$ and all $p \in A$.

Injective: Suppose $\delta_k(p) = \delta_k(q)$. Then $kp = kq \implies p = q$ (since $k \neq 0$).

Surjective: For any $q \in A$, set $p = q/k$. Then $\delta_k(p) = k \cdot (q/k) = q$.

Thus, δ_k is a bijection. To show it is an isometry, recall the upper half-plane weighted taxicab metric depends on the y -coordinate. Under dilation, both the x and y coordinates scale by k , so the height changes as well. The metric is not preserved unless we correct for the change in weight. Therefore, **dilation is not a general isometry of $\mathbb{H}_T^2$ unless one rescales the metric accordingly**. For purposes of this paper, we only regard dilations about the origin as isometries if the metric is adjusted appropriately.

2. Translation Map. The translation map is defined as

$$\Phi_a(p) = p + a,$$

for some fixed $a \in A$ and all $p \in A$.

Injective: Suppose $\Phi_a(p) = \Phi_a(q)$. Then $p + a = q + a \implies p = q$.

Surjective: For any $r \in A$, set $p = r - a$. Then $\Phi_a(p) = (r - a) + a = r$.

Thus, Φ_a is a bijection. However, **not all translations preserve distance in the upper half-plane weighted taxicab metric**, particularly when the vertical component is involved. For horizontal translations (i.e., $a = (a_1, 0)$), the y -coordinate is preserved, and the weight function (depending on height) remains unchanged along horizontal segments. These translations preserve the weighted metric. But for general translations with vertical displacement, the weight at each point changes, and so distance is not preserved. Therefore, only **horizontal translations** are isometries in $\mathbb{H}_T^2$.

3. Reflection Map. The reflection map about the vertical axis $x = c$ is defined as

$$\Psi(p_1, p_2) = (2c - p_1, p_2).$$

Injective: Suppose $\Psi(p) = \Psi(q)$. Then $(2c - p_1, p_2) = (2c - q_1, q_2) \implies p_1 = q_1, p_2 = q_2$.

Surjective: For any $q = (q_1, q_2) \in A$, let $p = (2c - q_1, q_2)$. Then $\Psi(p) = q$.

Thus, Ψ is a bijection. Reflections about vertical axes preserve the y -coordinate and reverse the x -coordinate across $x = c$. Because height

(and hence weight) is preserved and horizontal distances remain symmetric, the (weighted) taxicab metric is preserved. Hence, vertical reflections are isometries.

9. OBSERVATIONS AND HYPERBOLICITY IN THE UPPER HALF-PLANE METRIC

Recall our metric on the upper half-plane:

$$d(p, q) = \ln \left(\frac{m}{p_2} \right) + \ln \left(\frac{m}{q_2} \right) + \ln |q_1 - p_1|,$$

where

$$m = \max \left\{ p_2, q_2, \frac{1}{2}|q_1 - p_1| \right\}.$$

Gromov Product and δ -Hyperbolicity. For x, y, z, w in a metric space (X, d) , the Gromov product at z is

$$(x, y; z) = \frac{1}{2}(d(x, z) + d(y, z) - d(x, y)),$$

and (X, d) is δ -hyperbolic if

$$(x, y; z) \geq \min\{(x, w; z), (y, w; z)\} - \delta \quad \forall x, y, z, w.$$

10. GROMOV HYPERBOLICITY IN $\mathbb{H}_T^2$

Let $\mathbb{H} = \{(u_1, u_2) \in \mathbb{R}^2 : u_2 > 0\}$ and set

$$\mu(u, v) := \max\{u_2, v_2\} \quad (u = (u_1, u_2), v = (v_1, v_2)).$$

The distance and Gromov product are

$$(1) \quad d(u, v) = \ln \frac{\mu(u, v)}{u_2} + \ln \frac{\mu(u, v)}{v_2} + \frac{|u_1 - v_1|}{\mu(u, v)}.$$

$$(2) \quad (x \cdot y)_z = \frac{1}{2}(d(x, z) + d(y, z) - d(x, y)).$$

Substituting (1) into (2). With $x = (x_1, x_2)$, $y = (y_1, y_2)$, $z = (z_1, z_2)$,

$$\begin{aligned} G(x, y; z) &:= (x \cdot y)_z = \frac{1}{2}[d(x, z) + d(y, z) - d(x, y)] \\ &= \frac{1}{2} \left[\ln \frac{\mu(x, z)}{x_2} + \ln \frac{\mu(x, z)}{z_2} + \frac{|x_1 - z_1|}{\mu(x, z)} \right. \\ &\quad \left. + \ln \frac{\mu(y, z)}{y_2} + \ln \frac{\mu(y, z)}{z_2} + \frac{|y_1 - z_1|}{\mu(y, z)} \right. \\ &\quad \left. - \ln \frac{\mu(x, y)}{x_2} - \ln \frac{\mu(x, y)}{y_2} - \frac{|x_1 - y_1|}{\mu(x, y)} \right]. \end{aligned}$$

Collecting logarithmic terms. Since $\ln a + \ln b = \ln(ab)$ and $-\ln a = \ln(1/a)$,

$$(\text{logs}) = 2 \ln \left(\frac{\mu(x, z) \mu(y, z)}{\mu(x, y) z_2} \right).$$

Final expression.

(3)

$$G(x, y; z) = \ln \left(\frac{\mu(x, z) \mu(y, z)}{\mu(x, y) z_2} \right) + \frac{1}{2} \left(\frac{|x_1 - z_1|}{\mu(x, z)} + \frac{|y_1 - z_1|}{\mu(y, z)} - \frac{|x_1 - y_1|}{\mu(x, y)} \right).$$

Lemma 10.1 (Four-point (Gromov) form). *A metric space (X, d) is δ -hyperbolic iff for all x, y, z, w ,*

$$G(x, y; w) \geq \min\{G(x, z; w), G(y, z; w)\} - \delta,$$

where $G(a, b; c) = \frac{1}{2}(d(a, c) + d(b, c) - d(a, b))$.

Lemma 10.2 (Elementary min-log identities). *For $a, b, c > 0$:*

- (i) $\min\{a + c, b + c\} = \min\{a, b\} + c$
- (ii) $\min\{ac, bc\} = c \min\{a, b\}$
- (iii) $\min\{\frac{a}{c}, \frac{b}{c}\} = \frac{\min\{a, b\}}{c}$
- (iv) $\min\{\ln a, \ln b\} = \ln(\min\{a, b\})$
- (v) $\min\{\frac{1}{a}, \frac{1}{b}\} = \frac{1}{\max\{a, b\}}$

Bounding the four-point difference. Define

$$f(x, y, z, w) := \min\{G(x, z; w), G(y, z; w)\} - G(x, y; w).$$

Using (3), write

$$G(x, z; w) = \ln A + T_{xz}, \quad G(y, z; w) = \ln B + T_{yz}, \quad G(x, y; w) = \ln C + T_{xy},$$

where

$$A = \frac{\mu(x, w) \mu(z, w)}{\mu(x, z) w_2}, \quad B = \frac{\mu(y, w) \mu(z, w)}{\mu(y, z) w_2}, \quad C = \frac{\mu(x, w) \mu(y, w)}{\mu(x, y) w_2},$$

and

$$T_{ab} := \frac{1}{2} \left(\frac{|a_1 - w_1|}{\mu(a, w)} + \frac{|b_1 - w_1|}{\mu(b, w)} - \frac{|a_1 - b_1|}{\mu(a, b)} \right).$$

A crude estimate (via chained triangle inequalities) gives $-1 \leq T_{ab} \leq 2$.

Thus,

$$\frac{\min\{A, B\}}{C} = \frac{\mu(x, y) \mu(z, w)}{\mu(x, w) \mu(y, w)} \min \left\{ \frac{\mu(x, w)}{\mu(x, z)}, \frac{\mu(y, w)}{\mu(y, z)} \right\}.$$

$$f(x, y, z, w) \leq \ln \left(\frac{\min \left\{ \frac{\mu(x, w)}{\mu(x, z)}, \frac{\mu(y, w)}{\mu(y, z)} \right\} \cdot \mu(z, w) \cdot \mu(x, y)}{\mu(x, w) \mu(y, w)} \right) + 3$$

11. LUKA'S SECTION

11.1. Finding isometries.

Motivation: In an attempt to determine which maps are isometries for the upper half plane model in taxicab space, we explore such maps that preserve norms. This work is done separate from using a distance function to establish isometries. We first start with a map Ψ that reflects a point p horizontally. More is said on the bijectivity of these maps in Kevin's section.

Notation for finding isometries: For any isometry Ψ we define the push-forward map at a point $p \in \mathbb{H}_T^2$ given by $p = (p_1, p_2)$ as $\Psi_* : T_p(\mathbb{H}_T^2) \rightarrow T_{\Psi(p)}(\mathbb{H}_T^2)$. The vector $\Psi_*(v)$ can be calculated with a Jacobian matrix on the tangent space:

$$\Psi_*(v) = \begin{bmatrix} \frac{\partial \psi_1}{\partial p_1} & \frac{\partial \psi_1}{\partial p_2} \\ \frac{\partial \psi_2}{\partial p_1} & \frac{\partial \psi_2}{\partial p_2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

We aim to show that for any vector v in the tangent space of a point p , the vector has the same norm as the push-forward vector $\Psi_*(v)$ at $\Psi(p)$.

$$\|\Psi_*(v)\|_{\Psi(p)} = \|v\|_p.$$

Horizontal reflection map Ψ : We define a horizontal reflection map $\Psi : \mathbb{H}_T^2 \rightarrow \mathbb{H}_T^2$, where a point p after the mapping becomes $\Psi(p) = (-p_1, p_2)$.

$$\Psi_*(v) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -v_1 \\ v_2 \end{bmatrix}$$

Taking the norm of $\Psi_*(v)$ using the taxicab metric gives

$$\|\Psi_*(v)\| = |-v_1| + |v_2| = |v_1| + |v_2| = \|v\|.$$

With respect to a point p in the upper half-plane model:

$$\|v\|_p = \frac{1}{p_2}(|v_1| + |v_2|).$$

After mapping, we have push-forward vector with respect to $\Psi_{(p)}$.

$$\|\Psi_*(v)\|_{\Psi(p)} = \frac{1}{\Psi_2(p)}(|v_1| + |v_2|) = \frac{1}{p_2}(|v_1| + |v_2|) = \|v\|_p$$

Dilation map δ : We define a dilation centered about the origin $\delta : \mathbb{H}_T^2 \rightarrow \mathbb{H}_T^2$, where a point p after mapping becomes $\delta(p) = (kp_1, kp_2)$. The push-forward vector evaluates to a scalar multiple on v :

$$\delta_*(v) = (k)(I_n)v = kv.$$

Taking the norm of the push-forward vector at a point p :

$$\|\delta_*(v)\|_{\delta(p)} = \frac{1}{\delta_2(p)}(k|v_1| + k|v_2|) = \frac{1}{kp_2} \cdot k \cdot (|v_1| + |v_2|) = \|v\|_p.$$

Horizontal translation map Φ : We define a horizontal translation map $\Phi : \mathbb{H}_T^2 \rightarrow \mathbb{H}_T^2$ where the point p after mapping becomes $\Phi(p) = (p_1 + a, p_2)$. The push-forward evaluates to just v , so taking the norm of the push-forward vector at a point p gives:

$$\|\Phi_*(v)\| = \frac{1}{\Phi_2(p)}(|v_1| + |v_2|) = \frac{1}{p_2}(|v_1| + |v_2|) = \|v\|_p.$$

11.2. Finding η .

Motivation: In an attempt to find a length minimizing curve from p to q when $q_2 > p_2$, we use a curve η consisting of three line segments $\sigma_{p,a}, \sigma_{a,b}, \sigma_{b,q}$. The first line segment goes from p to $a = (p_1, (q_2 + h) - p_2)$ then from a to $b = (q_1, q_2 + h)$, then back to q . We can establish the length of η by adding up the lengths of the segments.

$$\mathcal{L}(\sigma_{p,a}) = \ln\left(\frac{q_2 + h}{p_2}\right) \quad \mathcal{L}(\sigma_{a,b}) = \frac{|q_1 - p_1|}{q_2 + h} \quad \mathcal{L}(\sigma_{b,q}) = \ln\left(\frac{q_2 + h}{q_2}\right)$$

$$\mathcal{L}(\eta) = \mathcal{L}(\sigma_{p,a}) + \mathcal{L}(\sigma_{a,b}) + \mathcal{L}(\sigma_{b,q})$$

$$\mathcal{L}(\eta) = \ln\left(\frac{(q_2 + h)^2}{q_2 p_2}\right) + \frac{|q_1 - p_1|}{q_2 + h}$$

Taking the derivative with respect to h , and setting it equal to zero gives the local minima for an optimized h value:

$$\frac{d\mathcal{L}}{dh} = \frac{2}{q_2 + h} - \frac{|q_1 - p_1|}{(q_2 + h)^2}$$

$$0 = \frac{2}{q_2 + h} - \frac{|q_1 - p_1|}{(q_2 + h)^2}$$

$$|q_1 - p_1| = \frac{2}{q_2 + h} \cdot (q_2 + h)^2 = 2(q_2 + h) \Rightarrow h^* = \frac{|q_1 - p_1|}{2} - q_2$$

We have found an optimal value given by h^* when $h^* > 0$. Next, we can plug in h^* to find the shortest distance η curve. See Kevin's section for analysis of the second derivative and the inflection point. For ease of calculation, let $d = |q_1 - p_1|$.

$$\begin{aligned} \mathcal{L}_{\min} &= \ln \left(\frac{(q_2 + h^*)^2}{q_2 p_2} \right) + \frac{|q_1 - p_1|}{q_2 + h^*} \\ &= \ln \left(\frac{d^2/4}{p_2 q_2} \right) + \frac{d}{d/2} \\ &= \ln d^2 - \ln(4p_2 q_2) + 2 = \mathcal{L}_{\min}, \quad \text{if } h^* > 0. \end{aligned}$$

Conclusion: We found certain " η " curves with a parameter h that extend upwards from p to $q_2 + h$, horizontally above q and back down to q . The shortest distance η curve is given by $\mathcal{L}_{\min}$. Alina's section covers the use of eta curves in establishing a distance function.

11.3. Gromov Hyperbolicity.

11.3.1. Motivation. In $\mathbb{H}_T^2$, an estimate for the hyperbolicity of the space using Gaussian curvature becomes a challenge given the lack of (inner products and norms). For that reason, we use Mikhail Gromov's findings about (non-euclidean) hyperbolic space to show that the space is indeed hyperbolic.

Gromov product and distances: We begin by using the Gromov Product to measure (travel length) of three points. Then, we move to the Gromov Hyperbolicity inequality which gives a measure of " δ -hyperbolicity" (given 4 points with geodesics connecting them). The Gromov Product equation is defined as so:

$$G(x, y; z) = \frac{1}{2}[d(x, z) + d(y, z) - d(x, y)]$$

Now we can rewrite the equation using the distance function with μ as defined earlier in the $\mathbb{H}_T^2$ distance function section.

$$\begin{aligned} G(x, y; z) = & \frac{1}{2} \left[\left(\ln\left(\frac{\mu(x, z)}{x_2}\right) + \ln\left(\frac{\mu(x, z)}{z_2}\right) + \frac{1}{\mu(x, z)}|x_1 - z_1| \right) \right. \\ & + \left(\ln\left(\frac{\mu(y, z)}{y_2}\right) + \ln\left(\frac{\mu(y, z)}{z_2}\right) + \frac{1}{\mu(y, z)}|y_1 - z_1| \right) \\ & \left. - \left(\ln\left(\frac{\mu(x, y)}{x_2}\right) + \ln\left(\frac{\mu(x, y)}{y_2}\right) + \frac{1}{\mu(x, y)}|x_1 - y_1| \right) \right] \end{aligned}$$

Combining terms, we get the resulting equation:

$$(1) \quad G(x, y; z) = \ln \left(\frac{\mu(x, z) \cdot \mu(y, z)}{\mu(x, y) \cdot z_2} \right) + \frac{1}{2} \left[\frac{|x_1 - z_1|}{\mu(x, z)} + \frac{|y_1 - z_1|}{\mu(y, z)} - \frac{|x_1 - y_1|}{\mu(x, y)} \right]$$

δ -hyperbolicity: The Gromov Hyperbolicity inequality says that a space is δ -hyperbolic if there exists a δ for all points x, y, z, w , the inequality holds:

$$G(x, y; w) \geq \min \{G(x, z; w), G(y, z; w)\} - \delta$$

We can rewrite the equation and move δ to one side, and let there be a function f be defined as so, such that $\delta \geq f$.

$$f(x, y, z, w) = \min \{G(x, z; w), G(y, z; w)\} - G(x, y; w)$$

Our goal is to show that f is bounded, because this means that there is an upper bound on δ , and our space is hyperbolic.

We can simplify the minimum function inside f by using information from Equation (1). The inside of the natural log term can be given as A , and the right side of this equation contains three terms including the $\frac{1}{2}$ that we can call T . T is bounded from $-1 \leq T \leq 2$, so we will use the upper bound to further simplify.

$$\begin{aligned} \min \{G(x, z; w), G(y, z; w)\} &= \min \{\ln(A) + T_{x,y,z}, \ln(A) + T_{y,z,w}\} \\ &\leq \min \{\ln(A) + 2, \ln(A) + 2\} \\ &= \min \{\ln(A), \ln(A)\} + 2 \end{aligned}$$

The other Gromov Product term in f can be simplified to $G(x, y; w) \geq \ln(A) - 1$. Now we can rewrite f .

$$\begin{aligned} f(x, y, z, w) &\leq \min \{ \ln(A), \ln(B) \} + 2 - \ln(C) + 1 \\ &= \min \{ \ln(A), \ln(B) \} - \ln(C) + 3 \\ &= \ln(\min \{ (A), (B) \}) - \ln(C) + 3 \\ &= \ln\left(\frac{\min \{ (A), (B) \}}{(C)}\right) + 3 \end{aligned}$$

In an attempt to find an upper bound for f , we can now re-substitute the values inside the natural log terms.

$$f \leq \ln \left[\frac{\min \left\{ \frac{\mu(x, w) \cdot \mu(z, w)}{\mu(x, z) \cdot w_2}, \frac{\mu(y, w) \cdot \mu(z, w)}{\mu(y, z) \cdot w_2} \right\}}{\frac{\mu(x, w) \cdot \mu(z, w)}{\mu(x, z) \cdot w_2}} \right] + 3$$

Further algebraic rearranging of terms and max and min functions results in the natural log term containing a maximum function in the numerator and denominator. We aim to compare the cases in the numerator and denominator to achieve an upper bound for all of the terms in the natural log. The numerator contains these nine terms, in three sections.

- (1) $x_2 z_2, x_2 w_2, y_2 z_2, y_2 w_2$
- (2) $\frac{x_2}{2} |z_1 - w_1|, \frac{y_2}{2} |z_1 - w_1|, \frac{z_2}{2} |x_1 - y_1|, \frac{w_2}{2} |x_1 - y_1|$
- (3) $\frac{1}{4} |x_1 - y_1| \cdot |z_1 - w_1|$

The denominator contains some of the same terms, but has 18 overall, in three sections similar to the numerator.

- (1) $x_2 y_2, x_2 w_2, z_2 y_2, z_2 w_2, y_2 x_2, y_2 w_2, z_2 x_2, z_2 w_2,$
- (2) $\frac{x_2}{2} |y_1 - w_1|, \frac{z_2}{2} |y_1 - w_1|, \frac{y_2}{2} |x_1 - z_1|, \frac{w_2}{2} |x_1 - z_1|$
 $\frac{y_2}{2} |x_1 - w_1|, \frac{z_2}{2} |x_1 - w_1|, \frac{x_2}{2} |y_1 - z_1|, \frac{w_2}{2} |y_1 - z_1|$
- (3) $\frac{1}{4} |x_1 - z_1| \cdot |y_1 - w_1|, \frac{1}{4} |y_1 - z_1| \cdot |x_1 - w_1|$

We use cases to compare numerator and denominator combinations, in an attempt to find an upper bound for the fraction inside the natural log, to then find an upper bound for f to show this is a hyperbolic space.

Case 1: We take the maximum term in the numerator to be any term from section (1). If the denominator has a maximum

that is any term in (1) besides itself, the denominator is larger than the numerator, meaning our term from (1) will be less than or equal to 1.

Case 2: We take a term from (2) in the numerator and compare it with a (2) term in the denominator. Specifically, we take a term in the numerator with the same coefficient, like $\frac{x_2}{2}|z_1 - w_1|$ matching the terms in the denominator $\frac{x_2}{2}|y_1 - w_1|$, or $\frac{x_2}{2}|y_1 - z_1|$. This reduces the denominator to a maximum function with two cases. Recall the triangle inequality, and that the absolute value terms measure distance. We can then say that for our y, z, w terms:

$$\begin{aligned} |z_1 - w_1| &\leq |y_1 - z_1| + |y_1 - w_1| \\ &\leq 2 \max\{|y_1 - z_1|, |y_1 - w_1|\} \end{aligned}$$

This shows that this case is always less than or equal to 2. The other cases for the other numerator combinations in (2) are left to the reader?

Case 3: Finally for the last case we analyze the numerator term in (3), and the denominator combinations. Again, the goal is to find an upper bound for the fraction inside the natural log, to then find an upper bound for f . We start by taking the term $\frac{1}{4}|x_1 - y_1||z_1 - w_1|$ and breaking it up into constituent parts using the triangle inequality in Case 2. (More on the trapezoidal proof?) (also explain where the $1/4$ term went)

$$\begin{aligned} |x_1 - y_1| &\leq 2 \max\{|y_1 - z_1|, |x_1 - z_1|\} \\ |x_1 - y_1| &\leq 2 \max\{|x_1 - w_1|, |x_1 - w_1|\} \\ |z_1 - w_1| &\leq 2 \max\{|z_1 - y_1|, |w_1 - y_1|\} \\ |z_1 - w_1| &\leq 2 \max\{|z_1 - x_1|, |w_1 - x_1|\} \end{aligned}$$

There are more individual distance terms we can take from in (3), the denominator: $\{|x_1 - z_1|, |x_1 - w_1|, |y_1 - z_1|, |y_1 - w_1|\}$. Of all of these terms, one has to be the smallest. If we assume that $|x_1 - z_1|$ is the smallest term, then referring to the constituent parts above gives us:

$$\begin{aligned} |x_1 - y_1| &\leq 2|y_1 - z_1| \\ |z_1 - w_1| &\leq 2|w_1 - x_1| \end{aligned}$$

Multiplying these inequalities together gives:

$$\begin{aligned} |x_1 - y_1||z_1 - w_1| &\leq 4|y_1 - z_1||w_1 - x_1| \\ &= \frac{|x_1 - y_1||z_1 - w_1|}{|y_1 - z_1||w_1 - x_1|} \leq 4 \end{aligned}$$

This shows that the highest possible value for the fraction inside the natural log term is 4, and the function f can be expressed as $f \leq \ln(4) + 3$.

Conclusion: We have therefore found an upper bound for the equation f to be $\ln(4) + 3$, showing that $\mathbb{H}_T^2$ is hyperbolic.